

# Noah Farr

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## EDUCATION

### TU Darmstadt

*M.Sc. Artificial Intelligence and Machine Learning, GPA 1.12*

Darmstadt, Germany

*Oct. 2024 – Present*

### TU Darmstadt

*B.Sc. Computer Science, GPA 1.68*

Darmstadt, Germany

*Oct. 2020 – Sep. 2024*

## PUBLICATIONS

### Streaming Reinforcement Learning under Partial Observability with Real-Time Recurrent Learning

May 2026

*Noah Farr, Aryaman Reddi, Carlo D'Eramo, Jan Peters*

*“RL in Big Worlds” Workshop, RLC*

- Proposed a streaming RL algorithm that learns online from a single stream of experience, without replay buffers or batched updates, under partial observability
- Leveraged recurrent trace units for exact real-time recurrent learning (RTRL) at linear time and memory cost, outperforming streaming baselines and matching batched training under tight compute budgets

## SCHOLARSHIPS & AWARDS

### Best Paper Award

May 2026

*“RL in Big Worlds” Workshop, RLC*

### Research-Oriented Master’s Scholarship

*Zuse School ELIZA*

Apr. 2025 – Present

*Darmstadt, Germany*

## EXPERIENCE

### Visiting Researcher

Jan. 2026 – Mar. 2026

*Softmax*

*San Francisco, USA*

- Researched zero-shot cooperation in multi-agent reinforcement learning (MARL), training agents to coordinate with unfamiliar partners without prior joint training
- Studied emergent cooperative behavior on the Alignment League Benchmark, a suite measuring organic alignment between interacting agents

### Student Research Assistant

Oct. 2024 – Present

*PEARL, TU Darmstadt*

*Darmstadt, Germany*

- Built a high-fidelity digital twin of a robotic kitchen in NVIDIA Isaac Sim, modeling its assets and physics to support sim-to-real transfer
- Designed and ran GPU-parallelized robotic manipulation experiments in Isaac Lab on the digital twin

### Research Intern

May 2024 – Sep. 2024

*TNO-ESI*

*Eindhoven, Netherlands*

- Developed a reinforcement learning pipeline for automated root-cause diagnosis of high-tech systems, improving fault-identification accuracy by 10% over the existing baseline

## PROJECTS

### RLX | [github.com/noahfarr/rlx](https://github.com/noahfarr/rlx)

Feb. 2024

- Built and open-sourced RLX, the first RL library for Apple’s MLX: end-to-end on-device training with vectorized environments exceeding 1M steps/second on Apple silicon, grown to 248 stars and 20 forks

### Memorax | [github.com/noahfarr/memorax](https://github.com/noahfarr/memorax)

Jan. 2026

- Built Memorax, a JAX library treating memory as a first-class citizen, snapping composable primitives into architectures like GTrXL and xLSTM across 12 sequence models and 15 environments

## LANGUAGES

German (Native), English (C1)